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Intelligent Well Monitoring: Redefining Efficiency and Sustainability in the Energy Industry

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Abstract

The oil and gas industry has been a cornerstone of global energy, driving economic growth. Currently, it faces significant challenges including volatile prices, rising costs, and increasing environmental regulations. As traditional reserves deplete, optimizing operations and enhancing asset performance becomes essential. To navigate these issues, companies are adopting advanced technologies like Automated Field Optimization, which aim to improve efficiency and sustainability in a rapidly changing market landscape.

This methodology utilizes algorithms, machine learning (ML), and real-time data analytics to enhance production efficiency and optimize hydrocarbon recovery while addressing potential issues. It supports decision-making and reduces costs, promoting economically viable and environmentally sustainable operations.

The client needed a fully autonomous solution for a remote well pad faced with difficult terrain, limited connectivity, and increased Health, Safety, and Environment (HSE) risks. The Intelligent Well Monitor (IWM) was developed for comprehensive digitalization in well pad management.

The IWM integrates essential systems like artificial lift, chemical injection, and virtual flow metering into a unified framework. This platform provides autonomous control and real-time operator visibility, reducing the need for on-site staff and significantly improving efficiency and safety while mitigating risks.

The IWM platform was developed rapidly to meet the client's stringent requirements, demonstrating effective project execution within a limited timeframe. A strong collaboration between offshore teams, comprising skilled developers and domain experts, played a crucial role in delivering a fully functional, high-performance product.

IWM's sophisticated capabilities enable seamless remote operations management, ensuring uninterrupted production and safeguarding essential assets. This achievement highlights the platform's capability to enhance operational reliability, sustainability, and safety, even under adverse conditions.

Moreover, IWM sets a new standard for autonomous oil field management by harnessing state-of-the-art technology and industry expertise. The platform addresses immediate operational needs and prepares for scalability in the Energy Industry. Its design prioritizes efficiency and effectiveness, allowing organizations to optimize resource use while minimizing risks. As industries increasingly focus on sustainability and

innovation, IWM is a leading solution integrating advanced operational strategies. This progress reflects an important shift in how oil field management is approached, promising to reshape practices and elevate industry standards, ultimately contributing positively to the sector's evolution.

Digitalization is shaping the future of the oil and gas sector, driving substantial changes. Companies are increasingly adopting digital technologies to enhance efficiency, optimize operations, and promote sustainability. This shift requires integrating various fields like machine learning and advanced data analytics. By collaborating across teams — engineering, operations, and IT — the industry is creating innovative solutions. Digitalization not only updates old processes but also enables the sector to adapt swiftly to changing energy needs.

Introduction

The oil and gas industry continues to face growing operational, economic, and environmental challenges. As global energy demand persists and conventional reserves mature, the need for efficient, reliable, and technologically advanced production systems becomes increasingly critical. In response, many operators are pursuing digital transformation strategies aimed at enhancing field autonomy, optimizing artificial lift performance, and reducing operational exposure, particularly in remote and logistically difficult regions. Ecuador's Amazon basin, a historically prolific but technically complex hydrocarbon province, exemplifies this need. The region presents unique challenges that include reservoir depletion, high gas-liquid ratios, water encroachment, difficult terrain, and limited connectivity—all of which exacerbate operational risks and costs, especially when relying on conventional field management approaches.

Electrical Submersible Pumps (ESPs) have been a cornerstone of artificial lift systems in Ecuador for decades, particularly in mature and marginal fields. Despite their ubiquity, traditional ESP deployments in the country are increasingly encountering limitations that compromise performance and run life. Declining reservoir pressures increased free gas fractions, and wellbore trajectory complexity have led to frequent gas lock events, inefficient pump operation, and premature equipment failures. These issues are compounded by logistical barriers, such as long mobilization times for workover rigs and the difficulty of accessing remote pads, often located deep within the Amazon rainforest. In this context, conventional manual operations pose significant limitations in terms of responsiveness, safety, and efficiency.

To address these challenges, various innovations in ESP design, chemical injection automation, and edge-based analytics have been trialed in Ecuador with promising results. For example, in four mature Ecuadorian fields, a fit-for-purpose ESP system integrated with extended gas-handling capabilities was deployed to manage high gas volume fractions and abrasive flow regimes, resulting in a 17% production uplift and a 50% reduction in ESP failures related to low productivity [3]. These developments have pushed the boundaries of ESP reliability in low-pressure environments and demonstrated the feasibility of more sophisticated artificial lift configurations. However, even these advancements still rely heavily on routine human intervention and centralized decision-making, both of which are subject to latency, error, and high exposure to Health, Safety, and Environment (HSE) risks.

The implementation of autonomous systems in oilfield operations offers a paradigm shift in how artificial lift, chemical dosing, instrumentation control, and well testing can be managed—particularly in remote and high-risk environments. Autonomous platforms reduce the reliance on field personnel by enabling real-time monitoring, control, and decision-making directly at the wellsite. These systems utilize edge computing, machine learning algorithms, and integrated sensing infrastructure to facilitate responsive and optimized operations. A notable advancement in this domain is the Intelligent Well Monitor (IWM) platform, which integrates multiple field systems into a single autonomous control environment, supporting artificial lift optimization, chemical injection, air compression control, and well testing with minimal human input.

Prior implementations of partial autonomy in ESP fields have demonstrated the value of real-time surveillance and control. One such example in Ecuador involved the deployment of a smart alarm suite using

an edge-based virtual flowmeter to monitor ESP operating conditions. This system utilized physics-based flow rate estimation and machine learning to proactively identify performance deviations and raise alarms under eight defined scenarios, including motor winding temperature fluctuations and emulsion formation [2]. The use of edge analytics in this context allowed for faster detection of anomalies, reduced false positives, and improved response times. It also exemplified how hybrid systems—merging physics models with machine learning—can improve both equipment protection and production outcomes. These advances also reduce carbon footprint and deferred production time, contributing to sustainability goals alongside operational objectives. However, that will be the subject of the subsequent paper in this series.

Another critical aspect of field autonomy in remote Ecuadorian assets is the ability to manage corrosion and scaling risks without manual intervention. One project implemented an autonomous corrosion and scale management system in ESP-lifted wells, utilizing edge analytics and chemical dosing algorithms to maintain optimal chemical injection rates based on real-time flow data and virtual scale/corrosion models [1]. This closed-loop control system allowed for precise, condition-based chemical injection, minimizing both over-treatment and under-protection. In environments where manual sampling and adjustment cycles may take weeks, the continuous protection offered by autonomous dosing substantially improves asset integrity and reduces workover frequency.

The integration of such systems into a cohesive IWM platform marks a significant leap in operational maturity. In the Ecuadorian Amazon context, where connectivity is often unreliable and terrain challenges delay response times, autonomy ensures continuity and performance. The region's political and regulatory landscape further incentivizes these solutions. Ecuador's upstream sector has undergone several contractual reforms, transitioning from service contracts to more flexible joint venture and participation models. Under these models, operators bear greater responsibility for cost and performance optimization. Therefore, the economic rationale for automation and reduced OPEX exposure becomes even more compelling.

Moreover, the environmental scrutiny of oil operations in ecologically sensitive zones such as the Yasuni and surrounding areas has led to tighter HSE regulations and heightened social accountability. By reducing the frequency of on-site personnel visits, autonomous systems mitigate exposure risks and align better with corporate sustainability commitments. This is particularly important in operations where any unplanned event, such as a well failure or equipment breakdown, could take days to respond to due to geographic isolation.

Field digitalization initiatives in Ecuador have further revealed the latent potential of existing infrastructure. In one case, a mature asset with over 250 producing wells across seven reservoirs used digital choke modeling and cloud-based production tracking to identify bottlenecks and shortfall opportunities, reducing optimization workflows from several days to just hours [4]. This digital transformation created a solid foundation for subsequent automation, enabling engineers to focus on exception handling and value-added analysis rather than routine monitoring.

The integration of all these capabilities—real-time ESP health monitoring, autonomous chemical injection, high-efficiency artificial lift, and digital well testing—into a unified IWM represents the culmination of years of incremental innovation. It also represents a strategic response to the unique operational demands of Ecuador's oilfields. The implementation of IWM platforms transforms the operational model from reactive and manpower-intensive to predictive, proactive, and digitally scalable.

Challenge

The client in Ecuador had a well pad situated in the heart of the Amazon jungle that faced significant operational challenges that impede efficient production management and optimization. The remote location of the well pad results in prolonged response times for detecting and addressing production issues through manual operations, thereby exacerbating delays and operational inefficiencies. Moreover, the presence

of active communities and frequent indigenous social strikes further disrupts wellsite monitoring and operational processes, contributing to unpredictable shutdowns and operational halts.

Electrical power reliability is another critical issue, with frequent diesel power generation shutdowns leading to inconsistent power supply and operational disruptions. Flow assurance is compromised by high chemical demands required for treating emulsion, corrosion, and asphaltenes, indicating inefficiencies in chemical treatments and further complicating production processes. Additionally, the lack of well testing adversely impacts production reporting and allocation efficiency, making it difficult to accurately assess and optimize production output. Compounding these issues is the absence of surface measurement capabilities, which results in no operational monitoring or alarms for critical pressure and temperature conditions, thus hindering proactive management and timely intervention.

From an operational perspective, these challenges collectively contribute to significant production losses at the well pad. The lack of updated well tests and inefficient chemical treatment for flow assurance are particularly detrimental, as they hamper accurate production assessment and optimal resource allocation. The remote location further exacerbates these issues, highlighting the urgent need for innovative solutions to enhance monitoring, reliability, and efficiency in this challenging environment. The following sections will delineate the primary characteristics of the selected well pad and propose targeted solutions to address these operational challenges.

Proposed Solution

Intelligent Well Monitoring; the solution proposed to address these operational challenges is centered on digital transformation and integration of advanced digital solutions; a comprehensive solution involving autonomous management of the well pad throughout almost all operational parameters, managed through a single user interface, customized for the client's needs. The solution strategy was to incorporate siloed solutions that had been tried and tested in other various fields in Ecuador and around the globe that could be unified to provide comprehensive workflows.

This strategic approach aligns with the business objectives of the asset, aiming to enhance production, reduce downtimes, minimize CO₂ emissions, and optimize operational efficiencies. A key component of this solution is the real-time monitoring and surveillance of water injection and oil production, which is crucial for the success of the secondary recovery project. By reducing intervention rates, enabling early detection of events, and ensuring efficient system operation, the solution aims to prevent uncontrolled water channeling in the reservoir. Real-time detection of potential issues in water injection performance, coupled with improved process surveillance and data integration, will address the current limitations of manual operations, failure identification in waterflood flow regulator valves, and power generator fluctuations.

Additionally, the solution offers significant opportunities to optimize oil production through virtual flowmeters (VFM), compensating for the lack of well test data. This approach will provide accurate assessments of well potential and facilitate the identification of necessary debottlenecking processes, thereby avoiding production losses. The chemical injection system, currently managed by unreliable mechanical pumps, will benefit from enhanced surveillance and optimization. The continuity of chemical injection rates has been suboptimal, leading to scale deposits and corrosion in ESP wells, which result in failures and production shutdowns. Manual data acquisition and analysis delays the optimization of chemical injection rates, rendering treatments ineffective due to dynamic well conditions. The proposed solution leverages a collaborative digital cloud platform, incorporating Edge and cloud intelligence for real-time monitoring, ESP analysis, surface and downhole measurements, well tests, smart alarms, advanced workflows, and autonomous chemical injection. This comprehensive approach aims to maximize oil production, meet daily injection targets, support field development plans, enhance crew efficiency, reduce operating expenses (OPEX), and ultimately achieve the objectives of the waterflooding strategy while supporting CO₂ emission reductions.

The Intelligent Well Monitoring Solution

Project Oversight and Governance

Figure 1 provides a comprehensive overview of how the IWM solution was conceptualized, developed, overseen, and ultimately executed. The subsequent paragraphs elaborate on how each stage was executed.

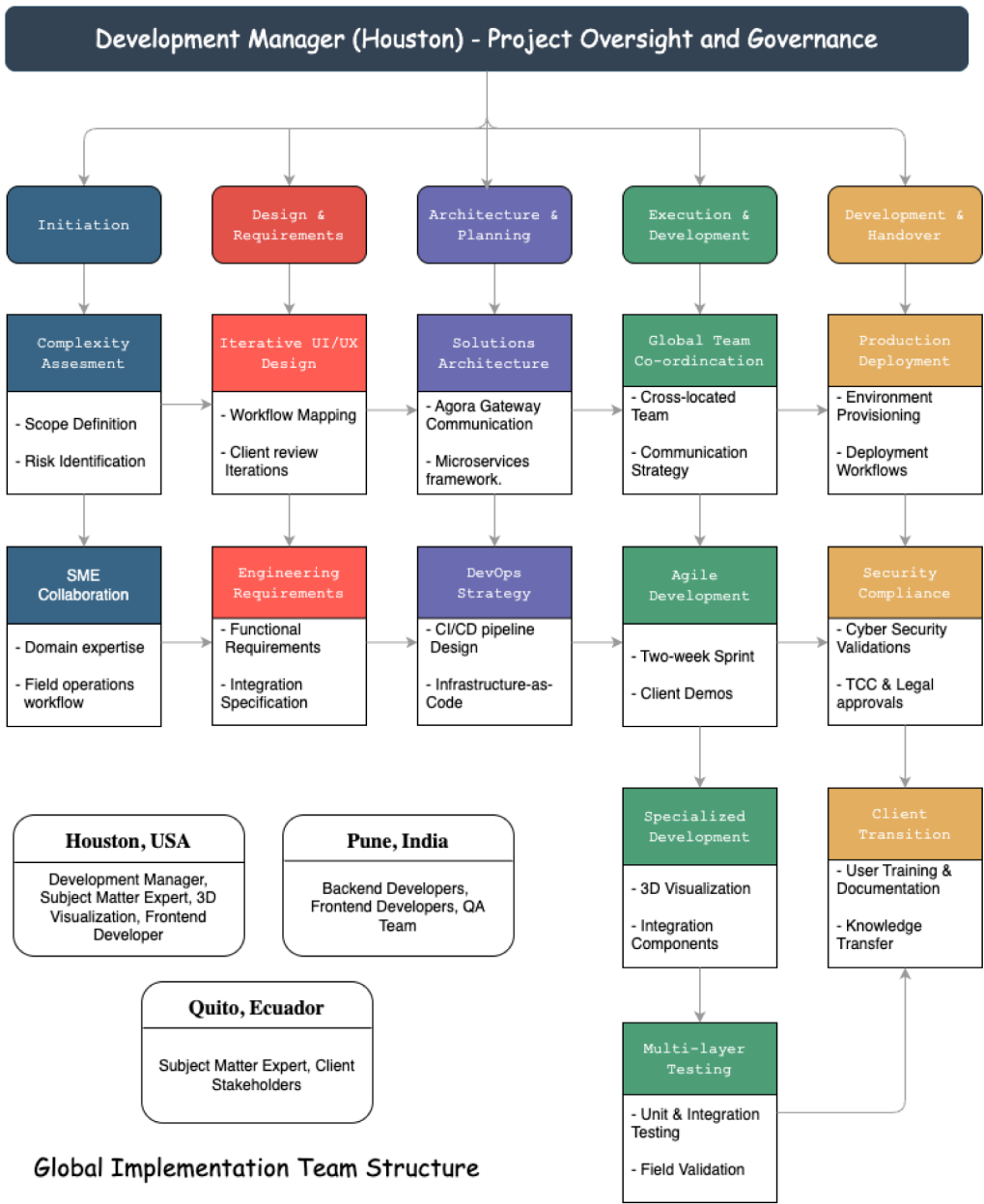


Figure 1—Intelligent Well Monitoring Solution Implementation and Governance Overview

Initial Assessment and Complexity Identification

The project initiation phase involved comprehensive assessment of scope and complexity factors. Early identification of potential challenges was crucial for developing appropriate mitigation strategies. Key complexity factors identified included:

- Complex workflows requiring specialized domain knowledge

- Geographically distributed teams operating across multiple time zones
- Specialized skill requirements, particularly in Industrial Internet of Things (IIoT) architectures and oil and gas domain expertise
- Integration requirements with existing third-party equipment
- Stringent client timelines driving aggressive delivery schedules

By acknowledging these complexities upfront, the project management approach was tailored to address potential bottlenecks and risks proactively rather than reactively.

Collaboration with Subject Matter Experts (SMEs) was established early in the project lifecycle to ensure deep understanding of field operations workflows. These SMEs provided critical domain knowledge regarding:

- Chemical injection processes and monitoring requirements
- ESP performance parameters and optimization techniques
- Various third-party equipment data formats and communication protocols
- Field operation procedures and user workflows

This collaboration proved invaluable in translating technical requirements into practical application features that aligned with actual field operations, reducing rework and ensuring the solution addressed verified operational needs.

Design and Requirements Engineering

The design phase employed an iterative approach involving close collaboration between SMEs, the UI/UX team, and client stakeholders. This process included:

- Initial workflow mapping based on SME input
- Development of preliminary UI/UX concepts
- Client review sessions to gather feedback
- Design refinement through multiple iterations
- Final design validation with end-users

The iterative approach ensured that the user interface accurately reflected operational workflows while maintaining usability and efficiency. Client involvement throughout this process fostered a sense of ownership and improved acceptance of the final product.

Following design finalization, comprehensive requirements were developed and documented in Azure DevOps, creating a single source of truth for the development team. Requirements were categorized into:

- Functional requirements detailing specific capabilities
- Non-functional requirements addressing performance, security, and scalability
- Integration requirements specifying interactions with third-party systems
- User interface requirements mapped to approved designs

Requirements were written with sufficient detail to guide development while maintaining traceability to business objectives. This structured approach to requirements management facilitated clear communication of expectations to the development team and provided a foundation for test case development.

Architectural Design and Infrastructure Planning

A comprehensive end-to-end architecture was developed to address both immediate project needs and future scalability requirements. Key architectural elements included:

- Agora Gateway for field equipment connectivity and edge computing capabilities
- Cloud services integration for data storage, analytics, and visualization
- Kubernetes-based infrastructure enabling cloud-agnostic deployment
- Microservices architecture supporting independent scaling of application components

The architecture prioritized scalability and reliability, allowing the solution to operate effectively across diverse cloud environments while maintaining consistent performance characteristics.

The architectural planning extended beyond system components to include deployment and operational considerations. This encompassed:

- Azure DevOps pipeline configurations for automated infrastructure provisioning
- Continuous integration and deployment workflows
- Infrastructure-as-Code templates for consistent environment replication
- Monitoring and alerting frameworks for operational oversight

This holistic architectural approach ensured that the solution could be reliably deployed, scaled, and maintained across development, testing, and production environments with minimal manual intervention.

Team Formation and Development Approach

A distributed development team was established with clear roles and responsibilities:

- Houston, USA
 - Project management team
 - Specialized resources for 3D visualization development
 - Frontend development team with Angular expertise (Partial team)
- Pune, India
 - Backend development team specializing in C# development
 - Frontend development team with Angular expertise (Partial team)
 - QA team with both manual and automation testing capabilities
- Quito, Ecuador
 - SMEs and client stakeholders

This geographic distribution presented communication challenges that were addressed through structured communication protocols, collaboration tools, and regular synchronization meetings, both in-person and virtual as required.

The project employed an Agile development methodology adapted to accommodate the distributed team structure:

- Two-week sprint cycles with defined delivery objectives
- Daily stand-up meetings scheduled to accommodate time zone differences
- Sprint planning and retrospective sessions with cross-location participation
- Continuous integration practices for early identification of integration issues
- Regular demo sessions with client stakeholders

This approach provided sufficient structure to coordinate distributed teams while maintaining the flexibility to adapt to changing requirements and priorities throughout the development lifecycle.

Project Execution Challenges and Solutions

An unexpected complexity emerged with the client's demands for specialized 3D visualization capabilities. This challenge was addressed by:

- Identifying specialized talent with required visualization expertise
- Integrating this resource into the development workflow
- Creating modular visualization components that could be integrated with the main application
- Establishing knowledge transfer sessions to build broader team understanding

This focused approach allowed the specialized visualization component to progress in parallel with core application development without creating dependencies that could delay overall progress.

The geographic distribution of teams across three continents required deliberate coordination strategies:

- Overlapping work hours were established to facilitate real-time collaboration
- Documentation standards were enhanced to reduce miscommunication risks
- Collaboration tools with asynchronous capabilities were implemented
- Regular status reports ensured alignment across all locations
- Periodic in-person and virtual meetings at critical project milestones were organized

These strategies enabled effective coordination despite significant time zone differences, ensuring that geography did not impede project progress.

Maintaining alignment with client expectations required structured communication:

- Biweekly meetings with clients to provide progress updates
- Application demonstrations at predetermined milestone, highlighting completed functionality
- Feedback collection and prioritization sessions
- Documentation of decisions and action items
- Transparent reporting of challenges and mitigation strategies

This regular communication cadence built trust with stakeholders while ensuring that development remained aligned with evolving business needs. The demonstration-focused approach made progress tangible to non-technical stakeholders and validated that the solution was meeting expectations.

Testing and Quality Assurance

A multi-layered testing approach was implemented to ensure application quality:

- Unit testing integrated with the development process
- Integration testing addressing component interactions
- System testing validating end-to-end workflows
- Performance testing under simulated load conditions
- Security testing to identify vulnerabilities
- User acceptance testing with actual end-users, both internal and external

This comprehensive approach identified issues at multiple levels, reducing the likelihood of critical defects reaching production.

A critical success factor was the implementation of field testing in actual operational environments:

- Development of a field-testing plan in collaboration with client stakeholders
- Securing necessary approvals for equipment connections
- On-site testing with field engineers to validate workflows
- Real-time issue resolution and adaptation
- Documentation of field test results and client sign-off

Field testing provided validation under authentic operational conditions, identifying usability and integration issues that might not be apparent in simulated environments. Working directly with field engineers ensured that the application supported practical operational needs rather than just meeting technical specifications.

Deployment and Handover

Following successful field testing, a comprehensive production deployment plan was developed addressing:

- Environment provisioning requirements
- Data migration procedures
- Deployment sequence and fallback procedures
- Performance monitoring configurations
- User access provisioning
- Post-deployment validation checks

This detailed planning reduced deployment risks and ensured a smooth transition to production operations.

Prior to production deployment, rigorous security and compliance validation was conducted:

- Collaboration with the Cybersecurity team for vulnerability assessment

- Penetration testing to identify potential security weaknesses
- Compliance verification against industry standards
- Trade Compliance & Customs (TCC) review and approval
- Legal team validation of compliance requirements

This multi-faceted validation ensured that the solution met all security and compliance requirements before being released to the client field operations.

The transition to client operations followed a structured handover process:

- Identification and provisioning of client users with appropriate access levels
- Development of operational documentation and training materials
- Training sessions for system administrators and end-users
- Establishment of support processes and escalation procedures
- Provision of comprehensive project documentation and reports
- Formal acceptance and client sign-off

This methodical handover ensured that client teams were prepared to utilize and maintain the system effectively after the project team's involvement concluded.

Field Results

The details pertaining to the field implementation of the IWM software are the subject of a subsequent SPE paper. However, following are the overall benefits delivered by the software.

- **Production Increase**
 - The project achieved a 7% improvement in oil production during the first year, resulting in an additional 21,000 barrels for the field.
- **Reduction in Workover Interventions**
 - The number of workover interventions was reduced, with the failure index dropping from 0.5 to 0.26, improving process and personnel efficiency.
 - **Improved ESP Monitoring and Control:** The implementation of intelligent ESP monitoring, smart alarms, and remote control allowed early detection and diagnosis of issues such as emulsions, gas locks, tubing leaks, and low pump efficiency, reducing the likelihood of failures that require workover interventions.
 - **Optimized Chemical Injection:** Remote and autonomous control of chemical injection ensured consistent and effective treatment, preventing flow assurance problems and pump clogging, which previously led to ESP failures and subsequent workovers.
 - **Faster Response to Issues:** The system enabled real-time monitoring and quicker response to operational problems, minimizing downtime and the need for manual interventions.
 - **Reduction in ESP Failures:** The combined effect of these digital solutions led to a significant reduction in ESP failures, which directly decreased the frequency of workover interventions.

- **CO2 Emissions Reduction**

- The project led to a significant reduction of 25 tons of CO2 emissions, attributed to fewer ESP failures, which led to fewer workover interventions and less equipment mobilization.
 - Improved remote monitoring and automation minimized the need for personnel travel and on-site activities, having a further contributing effect to lower CO2 output. Details on how the various workflows that contributed to CO2 reduction will be the subject of another subsequent SPE Paper.

- **Operational Efficiency**

- The automated workflows saved 40% of the time required for well testing, reducing the process from 10 hours to 6 hours with less than 10% deviation in results.

- **Smart Alarms Accuracy**

- The smart alarm system for ESP monitoring achieved 81% accuracy in detecting events and failures, supporting proactive maintenance and risk reduction.
- Ongoing research to develop enhanced models that will deliver accuracy at a rate higher than 90%, is ongoing.

- **ESP Run Life Improvement**

- ESP failures in the field were reduced by more than 50%, with only one failure reported in the year after implementation.

- **Economic Impact**

- The project delivered a positive net present value (NPV), with a payback period estimated at three years post-implementation.

- **Client Benefits**

- The client can now remotely monitor and control chemical injection, improving response time and operational oversight, which was previously only possible through field visits.

Lessons Learned and Best Practices

The successful execution of this complex IIoT project yielded several valuable lessons and best practices:

- **Early SME engagement** is crucial for accurate requirements definition in specialized domains like oil and gas operations.
- **Iterative design approaches** with client involvement improve solution alignment with user needs and increase stakeholder buy-in.
- **Field testing in actual operational environments** reveals issues that may not be apparent in laboratory settings and builds client confidence.
- **Specialized skill integration** should be planned early when unique capabilities like 3D visualization are required.
- **Distributed team management** requires deliberate communication structures and collaboration tools to overcome geographic and time zone challenges.
- **Comprehensive architecture planning** should address both technical and operational aspects, including deployment automation and monitoring.
- **Security and compliance considerations** must be integrated throughout the development lifecycle rather than addressed as an afterthought.

- **Regular client demonstrations** maintain alignment with expectations and provide opportunities for course correction.
- **Documentation standardization** becomes increasingly important as team distribution increases.
- **Cross-functional coordination** between development, QA, cybersecurity, and legal teams requires deliberate orchestration to prevent bottlenecks.

Conclusion

This IIoT implementation for the oil and gas industry exemplifies effective project management in a technically complex, geographically distributed environment with stringent timeline constraints. The integration of Chemical Injection, ESP, and Multiphase Flow Measurement (MPFM) data from diverse sources into a unified platform required careful planning, execution, and stakeholder management.

Key success factors included early SME engagement, iterative design processes, comprehensive testing strategies including field validation, and meticulous deployment planning. Despite challenges such as geographic team distribution and specialized visualization requirements, the structured project management approach enabled successful delivery.

The deployment of an IWM system in a remote Ecuadorian well pad represents more than just technological evolution—it embodies a holistic approach to modernizing field operations in one of the world's most challenging oilfield environments. By drawing upon lessons learned from previous ESP deployments, autonomous chemical management systems, and edge-based analytics, this initiative sets a new standard for field autonomy. It directly addresses the pressing needs of mature field optimization, HSE risk reduction, and cost control, while also contributing to the long-term sustainability and digitalization goals of the operator.

The methodologies employed in this project demonstrate how appropriate planning, team organization, and stakeholder communication can overcome significant complexity factors in industrial technology implementations. By maintaining focus on both technical excellence and business alignment throughout the project lifecycle, the solution delivered genuine operational value while meeting security and compliance requirements.

The success of this project offers valuable insights for other regions facing similar challenges and serves as a reference model for future autonomous deployments across the global upstream sector.

Next Steps

A principal outcome of the implementation of the IWM software has been a significant reduction in field visits. This has resulted in a considerable reduction in CO₂ emissions (25 tons). This enhanced operational efficiency, while addressing critical HSE issues. The benefits were particularly pronounced during periods of political instability when physical access to the well pad was temporarily constrained. As mentioned earlier, the details of the various workflows that contributed to CO₂ reduction will be the subject of another SPE Paper. In addition, there are ongoing efforts to develop enhanced models that will deliver accuracy of the existing Smart Alarms, at a rate higher than 90%.

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